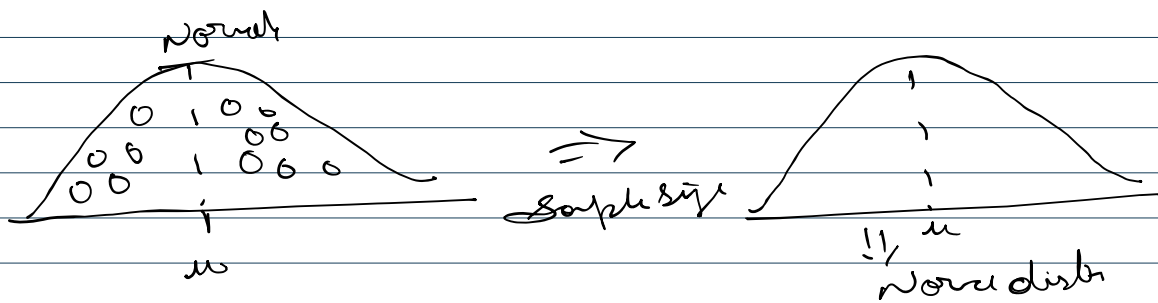
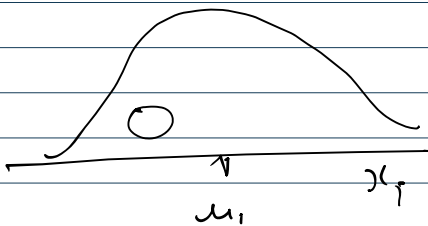




⇒ Sampling distribution will be normally distributed.

$$X \sim N(\mu, \sigma)$$

$$n = \text{sample size}$$



$$\begin{aligned} S_1 &= \{x_1, x_2, x_3, \dots, x_n\} = \overline{x_1} \\ S_2 &= \{x_1, x_2, \dots, x_n\} = \overline{x_2} \\ S_3 &= \{x_1, x_2, \dots, x_n\} = \overline{x_3} \\ &\vdots \\ S_n &= \dots \end{aligned}$$



$$n \geq 30$$

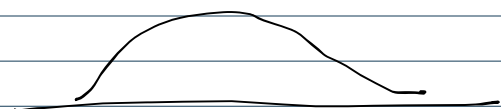
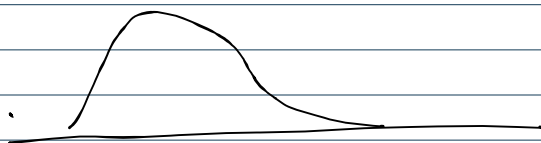
$n \rightarrow$  Sample Size should be  $\geq 30$

$$X \sim N(\mu, \sigma)$$

$$n \geq 30$$

$$S_1 = \overline{x_1}$$

$$S_2 = \overline{x_2}$$



$$X \sim N(\mu, \sigma)$$

$$X \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

